

ISyE6669 Homework 8

Fall 2025

1. Consider the following linear program:

$$\begin{array}{llllll}
 \min & -x_1 & - & 2x_2 & & \\
 \text{s.t.} & x_1 & - & x_2 & \leq & 1 \\
 & 2x_1 & - & x_2 & \geq & -1 \\
 & 2x_1 & + & x_2 & \leq & 5 \\
 & x_1 \geq 0, & x_2 \geq 0. & & &
 \end{array}$$

Graph the constraints of this linear program, and indicate the feasible region.

Solution: See Figure 1.

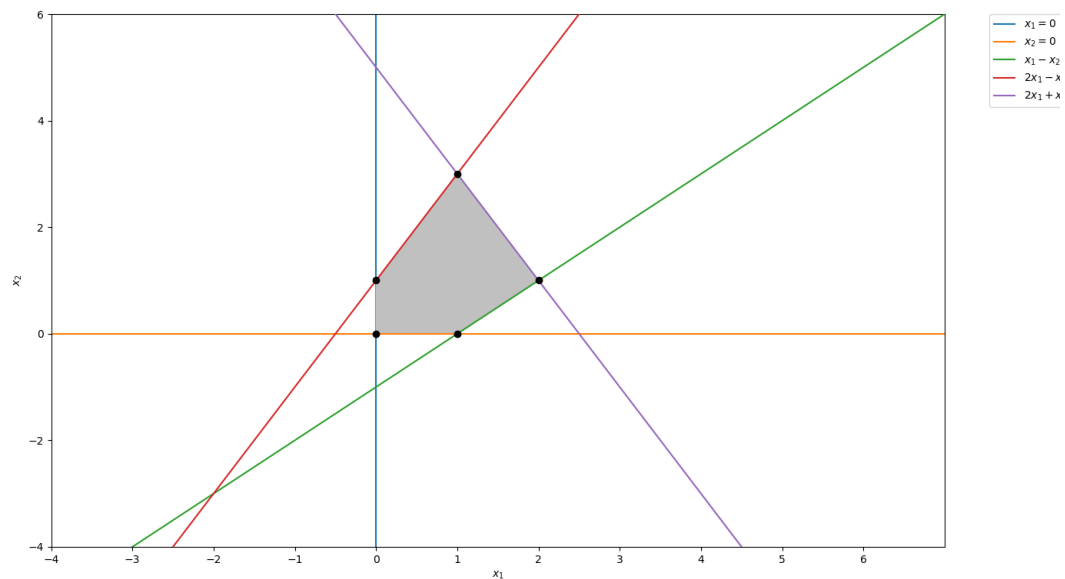


Figure 1: Feasible region of LP

2. To solve this problem using the simplex method, first transform it into a standard form LP.

Let x_3, x_4, x_5 be the slack variables with respect to the first, second, third constraint respectively. Denote $\mathbf{x} = [x_1, x_2, x_3, x_4, x_5]^\top$ as the vector of variables, and use the standard form notation:

$$\begin{aligned} \min \quad & \mathbf{c}^\top \mathbf{x} \\ \text{s.t.} \quad & \mathbf{Ax} = \mathbf{b} \\ & \mathbf{x} \geq \mathbf{0}, \end{aligned}$$

specify $\mathbf{c}, \mathbf{A}, \mathbf{b}$ for the above problem.

Solution: We can rewrite the problem as:

$$\begin{aligned} \min \quad & -x_1 - 2x_2 \\ \text{s.t.} \quad & x_1 - x_2 + x_3 = 1 \\ & 2x_1 - x_2 - x_4 = -1 \\ & 2x_1 + x_2 + x_5 = 5 \\ & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0, x_4 \geq 0, x_5 \geq 0. \end{aligned}$$

In this case, the $\mathbf{c}, \mathbf{A}$ and $\mathbf{b}$ are

$$\mathbf{c} = \begin{bmatrix} -1 \\ -2 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 1 & -1 & 1 & 0 & 0 \\ 2 & -1 & 0 & -1 & 0 \\ 2 & 1 & 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 1 \\ -1 \\ 5 \end{bmatrix}.$$

3. Now we want to solve the above standard-form linear program by the simplex method. If in an iteration of the Simplex method, there is any ambiguity about which nonbasic variable to enter the basis or which basic variable to exit the basis, use Bland's rule.

(a) For each iteration of the simplex method, write down the following information in the format given below:

- Iteration $k =$ numerical value, e.g. 1, 2, ...;
- Basis $\mathbf{B} = [\mathbf{A}_i, \mathbf{A}_j, \mathbf{A}_k] =$ numerical value (i.e. you need to specify i, j, k as well as the numerical values of the columns);
- Basis inverse $\mathbf{B}^{-1} =$ numerical value;
- Basic variable $\mathbf{x}_B = [x_i, x_j, x_k] =$ numerical value (you need to specify i, j, k and numerical values of x_i, x_j, x_k);
- Nonbasic variable $\mathbf{x}_N = [x_p, x_q] =$ numerical value (you need to specify p, q and numerical values of x_p, x_q);
- Reduced cost for each nonbasic variable $\bar{c}_p = c_p - \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{A}_p =$ numerical values; Same for $\bar{c}_q$; (you need to determine the index “?” for $\mathbf{A}_?$);
- Is the current solution optimal? If not, which nonbasic variable enters the basis?

- Direction $\mathbf{d}_B = -\mathbf{B}^{-1}\mathbf{A}_7 = \underline{\text{numerical value}}$. Does the simplex method terminate with an unbounded optimum?
- Min-ratio test $\theta^* = \min_{i:d_{B(i)} < 0} \{x_{B(i)}/(-d_{B(i)})\} = \min\{\underline{\text{numerical values of the ratios}}\} = \underline{\text{numerical value of } \theta^*}$.
- Which basic variable exits the basis?

Start at iteration $k = 1$ with the basis $\mathbf{B}^1 = [\mathbf{A}_3, \mathbf{A}_4, \mathbf{A}_5]$. Solve the above linear program with simplex method and write down all the information required above for each iteration. Also indicate the basic feasible solution at each step on the picture in (x_1, x_2) . What is the optimal solution of the above LP in $(x_1, \dots, x_5)$? What is the optimal cost?

From this exercise, you can see how the simplex method works and geometrically what each step is doing.

Solution:

Iteration $k = 1$

- $\mathbf{B} = [\mathbf{A}_3, \mathbf{A}_4, \mathbf{A}_5] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

- $\mathbf{B}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

- $\mathbf{x}_B = [x_3, x_4, x_5]^\top = [1, 1, 5]^\top$

- $\mathbf{x}_N = [x_1, x_2]^\top = [0, 0]^\top$

- Reduced cost for each nonbasic variable

$$\bar{c}_1 = c_1 - \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{A}_1 = -1$$

$$\bar{c}_2 = c_2 - \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{A}_2 = -2$$

- Both $c_1, c_2 < 0$ therefore the current solution is not optimal. Variable x_1 enters the basis according to Bland's rule.
- Direction $\mathbf{d}_B = -\mathbf{B}^{-1}\mathbf{A}_1 = [-1, 2, -2]^\top$. The simplex method does not terminate with an unbounded optimum.
- Min-ratio test $\theta^* = \min_{i:d_{B(i)} < 0} \{x_{B(i)}/(-d_{B(i)})\} = \min\{\frac{x_3}{-d_3}, \frac{x_5}{-d_5}\} = \min\{\frac{1}{1}, \frac{5}{2}\} = 1$.
- x_3 exits the basis

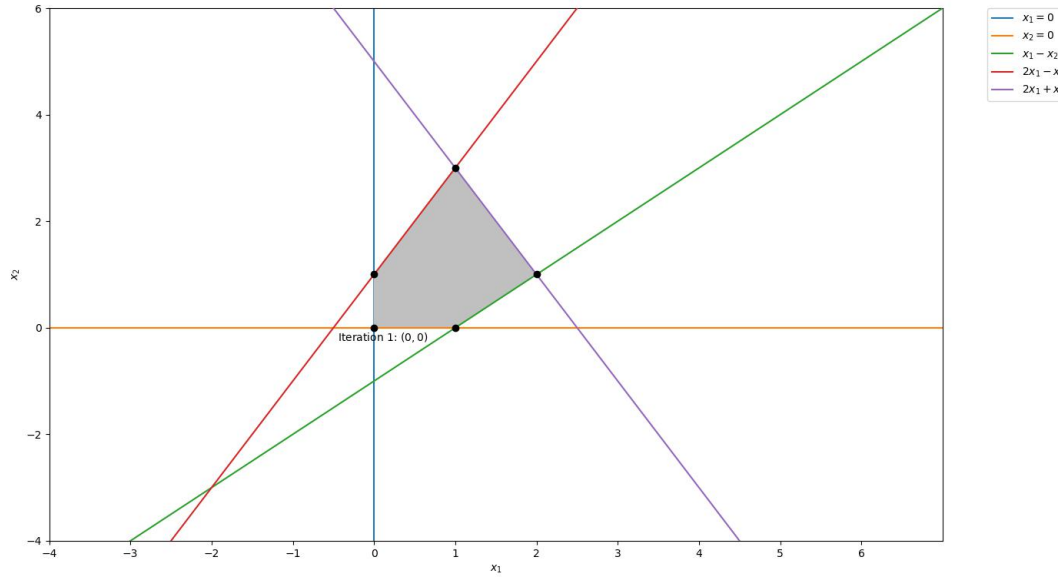


Figure 2: Iteration 1

Iteration $k = 2$

- $B = [A_1, A_4, A_5] = \begin{bmatrix} 1 & 0 & 0 \\ 2 & -1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$

- $B^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & -1 & 0 \\ -2 & 0 & 1 \end{bmatrix}$

- $x_B = [x_1, x_4, x_5]^\top = [1, 3, 3]^\top$

- $x_N = [x_2, x_3]^\top = [0, 0]^\top$

- Reduced cost for each nonbasic variable

$$\bar{c}_2 = c_2 - c_B^\top B^{-1} A_2 = -3$$

$$\bar{c}_3 = c_3 - c_B^\top B^{-1} A_3 = 1$$

- $c_2 < 0$ therefore the current solution is not optimal. Variable x_2 enters the basis.
- Direction $d_B = -B^{-1} A_2 = [1, 1, -3]^\top$. The simplex method does not terminate with an unbounded optimum.

- Min-ratio test $\theta^* = \min_{i:d_{B(i)} < 0} \{x_{B(i)}/(-d_{B(i)})\} = \min\{\frac{x_5}{-d_5}\} = \min\{\frac{3}{3}\} = 1$.
- x_5 exits the basis

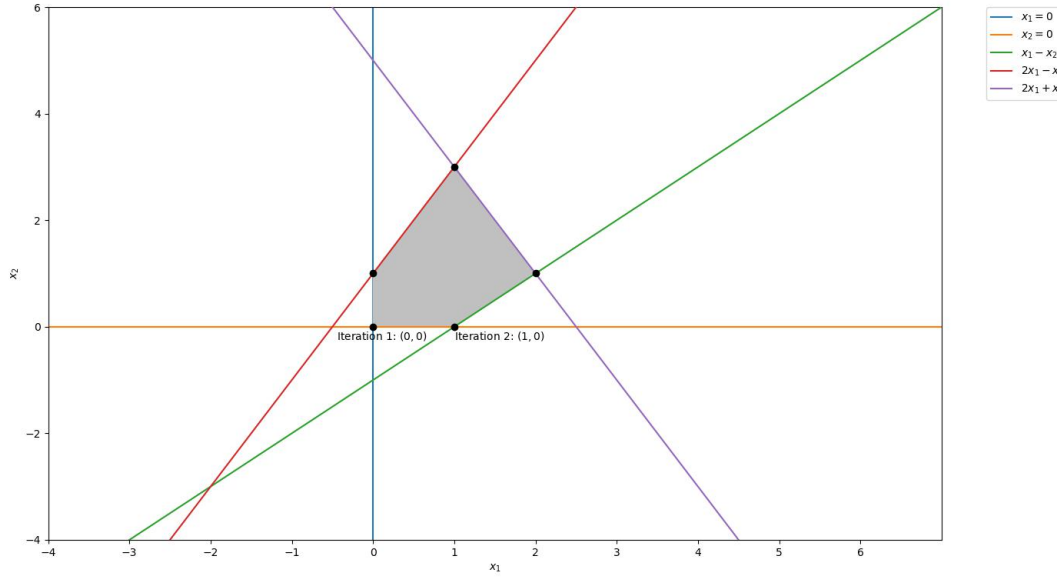


Figure 3: Iteration 2

Iteration $k = 3$

- $B = [A_1, A_2, A_4] = \begin{bmatrix} 1 & -1 & 0 \\ 2 & -1 & -1 \\ 2 & 1 & 0 \end{bmatrix}$
- $B^{-1} = \begin{bmatrix} 1/3 & 0 & 1/3 \\ -2/3 & 0 & 1/3 \\ 4/3 & -1 & 1/3 \end{bmatrix}$
- $x_B = [x_1, x_2, x_4]^\top = [2, 1, 4]^\top$
- $x_N = [x_3, x_5]^\top = [0, 0]^\top$
- Reduced cost for each nonbasic variable

$$\bar{c}_3 = c_3 - c_B^\top B^{-1} A_3 = -1$$

$$\bar{c}_5 = c_5 - c_B^\top B^{-1} A_5 = 1$$

- $c_3 < 0$ therefore the current solution is not optimal. Variable x_3 enters the basis.
- Direction $\mathbf{d}_B = -\mathbf{B}^{-1}\mathbf{A}_3 = [-1/3, 2/3, -4/3]^\top$. The simplex method does not terminate with an unbounded optimum.
- Min-ratio test $\theta^* = \min_{i:d_{B(i)} < 0} \{x_{B(i)}/(-d_{B(i)})\} = \min\{\frac{x_1}{-d_1}, \frac{x_4}{-d_4}\} = \min\{\frac{2}{1/3}, \frac{4}{4/3}\} = 3$.
- x_4 exits the basis

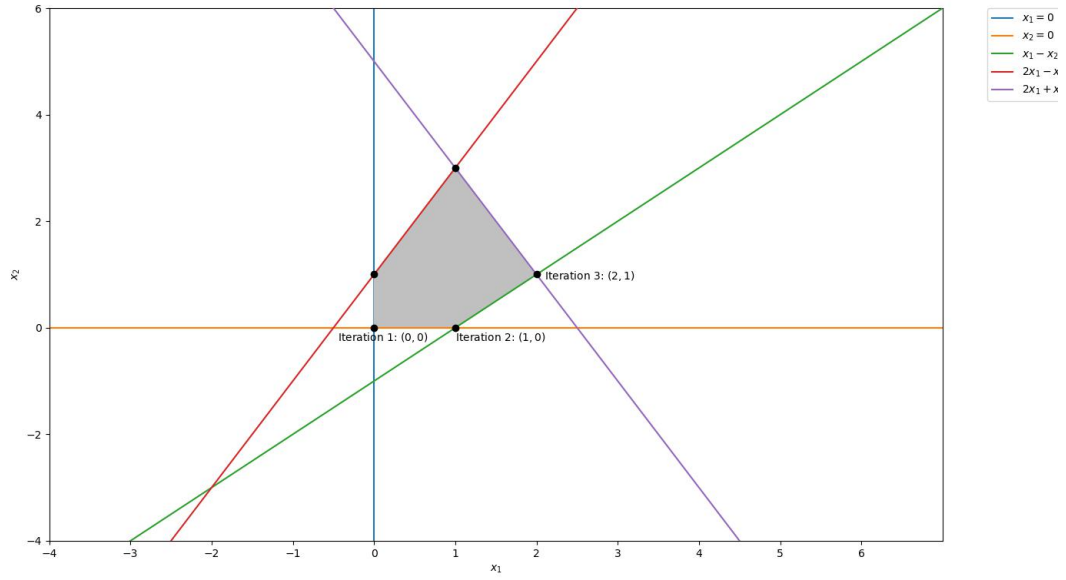


Figure 4: Iteration 3

Iteration $k = 4$

- $\mathbf{B} = [\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3] = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 2 & 1 & 0 \end{bmatrix}$
- $\mathbf{B}^{-1} = \begin{bmatrix} 0 & 1/4 & 1/4 \\ 0 & -1/2 & 1/2 \\ 1 & -3/4 & 1/4 \end{bmatrix}$
- $\mathbf{x}_B = [x_1, x_2, x_3]^\top = [1, 3, 3]^\top$
- $\mathbf{x}_N = [x_4, x_5]^\top = [0, 0]^\top$

- Reduced cost for each nonbasic variable

$$\bar{c}_4 = c_4 - \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{A}_4 = 0.75$$

$$\bar{c}_5 = c_5 - \mathbf{c}_B^\top \mathbf{B}^{-1} \mathbf{A}_5 = 1.25$$

- Both $c_4, c_5 > 0$ therefore the current solution is optimal.

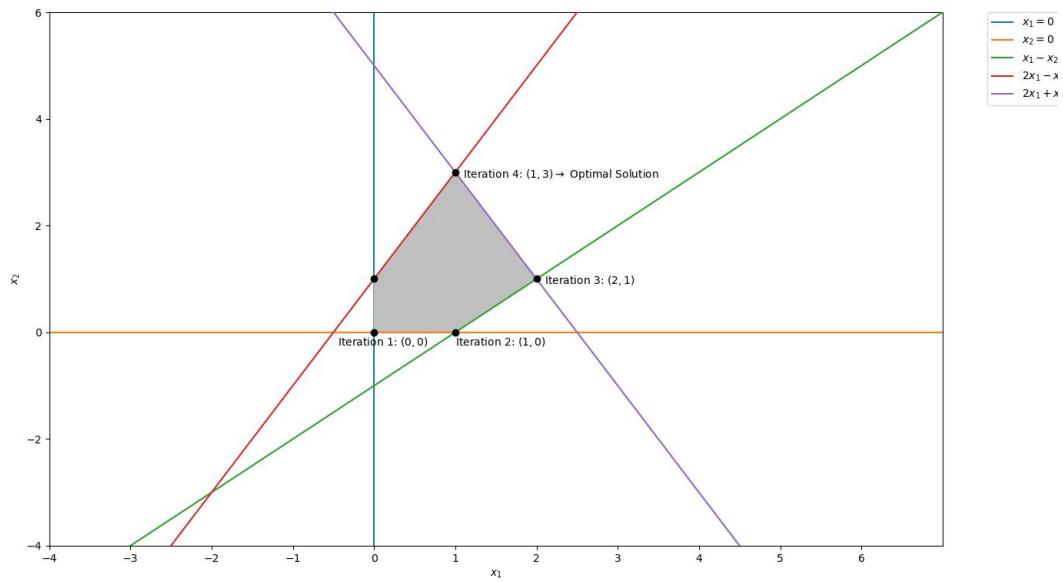


Figure 5: Iteration 4

Thus optimal solution is

$$\mathbf{x}^* = \begin{bmatrix} 1 \\ 3 \\ 3 \\ 0 \\ 0 \end{bmatrix}$$

and optimal cost is

$$\mathbf{c}^\top \mathbf{x}^* = [-1 \quad -2 \quad 0 \quad 0 \quad 0] \begin{bmatrix} 1 \\ 3 \\ 3 \\ 0 \\ 0 \end{bmatrix} = -7.$$